

CSIZG513 / ESZG513 / SSZG513: Network Security
Assignment #1 (+ Lab Activity #1)
(Total Marks: 10)

Sushmitha S 2026ht66027

- a. Every TLS record carries a ContentType value that identifies the higher-layer protocol using the record layer. Complete the table below. **[1 mark]**

ContentType value	Protocol it identifies
20	Change Cipher Spec
21	Alert
22	Handshake
23	Application Data

- b. Name the four upper-layer protocols that sit above the TLS Record Protocol in the TLS architecture. **[1 mark]**

Ans: Change Cipher Spec Protocol, Alert Protocol, Handshake protocol, Application Data Protocol

- c. The ClientHello and ServerHello messages each carry a Random (nonce) field. State its size in bytes and the two components it is made up of. **[1 mark]**

Ans: 32 Bytes. 4byte GMT Unix Time + 28 Byte random Value

- d. State what the master secret is used for, and name the three inputs from which it is derived. **[1 mark]**

Ans: Used to derive the keying material/session keys for the connection

Derived from pre-master secret + ClientHello.random + ServerHello.Random

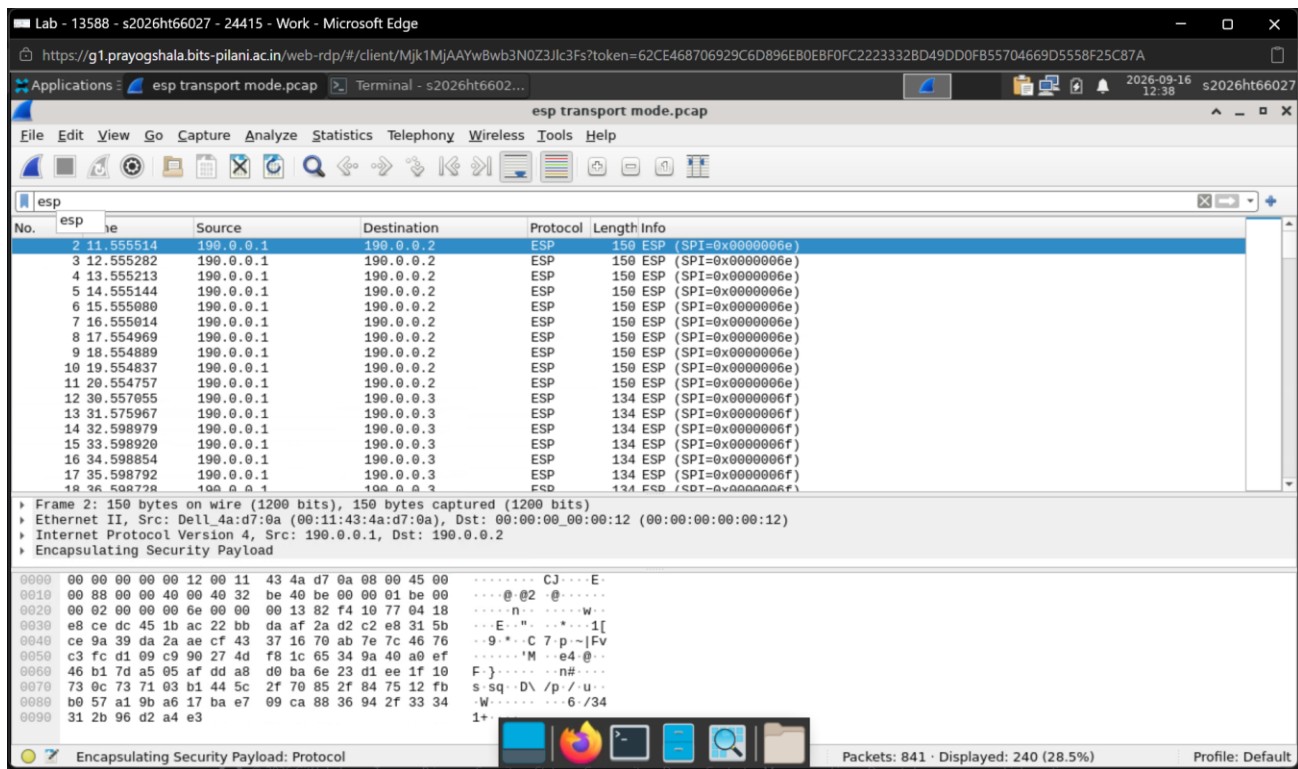
- e. TLS defines two alert levels. Name both, and state what happens to the connection in each case. **[1 mark]**

Ans: warning → connection may continue

Fatal → connection immediately terminated. Session can't continue.

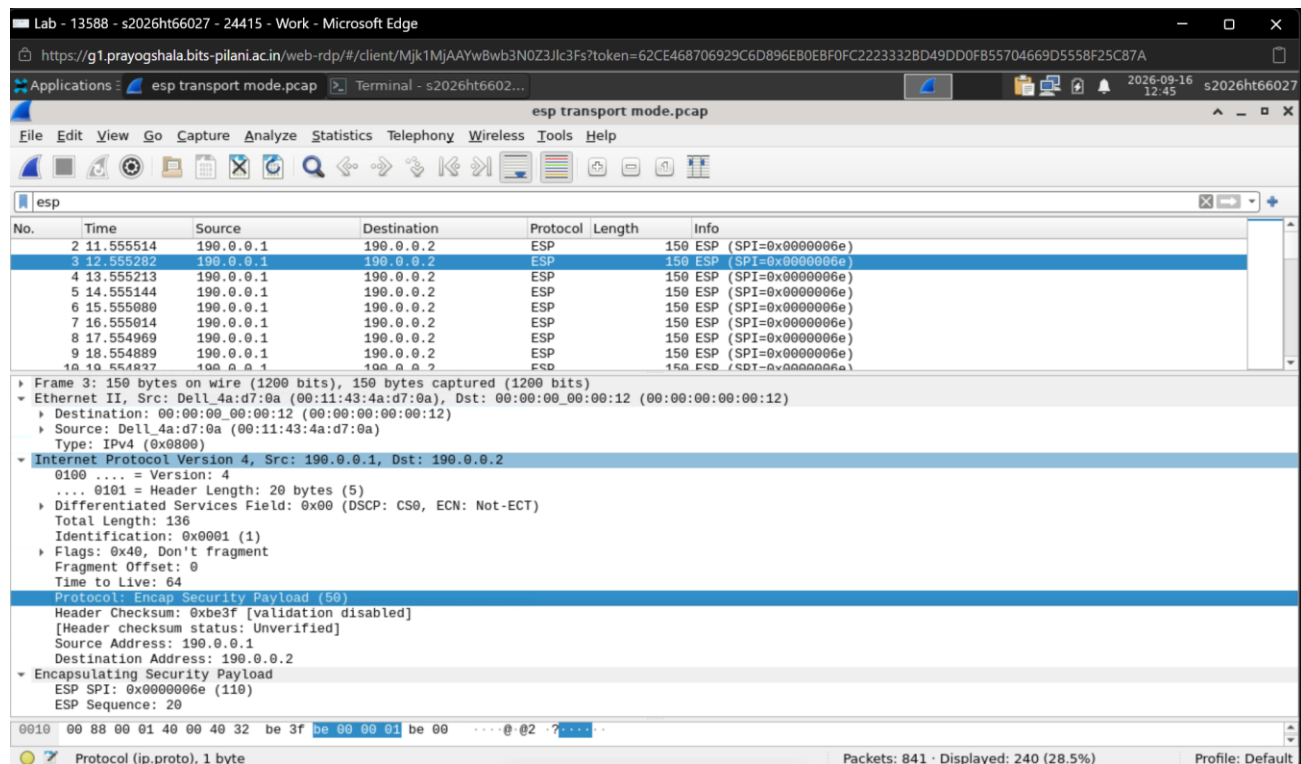
2. Use the Wireshark Network Protocol Analyzer to investigate IPsec-protected traffic and You can use packet safari as an option.

- a. Open the provided trace file "esp-transport-mode.pcap".
- c. Apply the display filter esp to show only the IPsec-protected packets.
- d. Attach a snapshot showing the packets displayed by this filter (as many as are visible in one window view). **[1 mark]**



e. Select any one ESP packet and expand both its Internet Protocol Version 4 and Encapsulating Security Payload sections, then answer the following:

i. What value appears in the Protocol field of the IPv4 header of these packets, and what does that value indicate? [1 mark]



Protocol field has a value of 50, IP payload is ESP

ii. Name the two fields that are visible in the ESP header, and give the value of each for the packet you selected. [1 mark]

Ans: two visible fields are Security Parameters Index and Sequence Number

iii. Compare several consecutive ESP packets sent to the same destination. State what you observe about one of the two fields above, and name the security service that this field supports. [1 mark]

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https://g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/Mjk1MjAAYwBwb3N0Z3Jlc3Fs?token=62CE468706929C6D896E80EBF0FC2223332BD49DD0FB5704669D5558F25C87A

Applications: esp transport mode.pcap Terminal - s2026ht66027...

esp transport mode.pcap

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esp

No.	Time	Source	Destination	Protocol	Length	Info
2	11.555514	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
3	12.555282	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
4	13.555213	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
5	14.555144	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
6	15.555080	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
7	16.555014	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
8	17.554969	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
9	18.554889	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
10	19.554827	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)

Frame 4: 150 bytes on wire (1200 bits), 150 bytes captured (1200 bits) on interface 0

Ethernet II, Src: Dell_4a:d7:0a (00:11:43:4a:d7:0a), Dst: 00:00:00:00:00:12 (00:00:00:00:00:12)

Destination: 00:00:00:00:00:12 (00:00:00:00:00:12)

Source: Dell_4a:d7:0a (00:11:43:4a:d7:0a)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 190.0.0.1, Dst: 190.0.0.2

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 136

Identification: 0x0002 (2)

Flags: 0x40, Don't fragment

Fragment Offset: 0

Time to Live: 64

Protocol: Encap Security Payload (50)

Header Checksum: 0xbe3e [validation disabled]

[Header checksum status: Unverified]

Source Address: 190.0.0.1

Destination Address: 190.0.0.2

Encapsulating Security Payload

ESP SPI: 0x0000000e (10)

ESP Sequence: 21

0010 00 88 00 02 40 00 40 32 be 3e be 00 00 01 be 00 ...@.02 ->.....

Protocol (ip.proto), 1 byte

Packets: 841 · Displayed: 240 (28.5%) Profile: Default

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https://g1.prayogshala.bits-pilani.ac.in/web-rdp/#/client/Mjk1MjAAYwBwb3N0Z3Jlc3Fs?token=62CE468706929C6D896E80EBF0FC2223332BD49DD0FB5704669D5558F25C87A

Applications: esp transport mode.pcap Terminal - s2026ht66027...

esp transport mode.pcap

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esp

No.	Time	Source	Destination	Protocol	Length	Info
2	11.555514	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)
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10	19.554827	190.0.0.1	190.0.0.2	ESP	150	ESP (SPI=0x0000000e)

Frame 5: 150 bytes on wire (1200 bits), 150 bytes captured (1200 bits) on interface 0

Ethernet II, Src: Dell_4a:d7:0a (00:11:43:4a:d7:0a), Dst: 00:00:00:00:00:12 (00:00:00:00:00:12)

Destination: 00:00:00:00:00:12 (00:00:00:00:00:12)

Source: Dell_4a:d7:0a (00:11:43:4a:d7:0a)

Type: IPv4 (0x0800)

Internet Protocol Version 4, Src: 190.0.0.1, Dst: 190.0.0.2

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 136

Identification: 0x0003 (3)

Flags: 0x40, Don't fragment

Fragment Offset: 0

Time to Live: 64

Protocol: Encap Security Payload (50)

Header Checksum: 0xbe3d [validation disabled]

[Header checksum status: Unverified]

Source Address: 190.0.0.1

Destination Address: 190.0.0.2

Encapsulating Security Payload

ESP SPI: 0x0000000e (10)

ESP Sequence: 22

0010 00 88 00 03 40 00 40 32 be 3d be 00 00 01 be 00 ...@.03 ->.....

Protocol (ip.proto), 1 byte

Packets: 841 · Displayed: 240 (28.5%) Profile: Default

Ans:

The ESP sequence increases for consecutive packets sent to the same destination. This field supports the anti-replay security service, helping detect and reject replayed packets.

iv. The same trace also contains ICMP echo requests sent in the clear to other destinations. State what this tells you about how the sender decides which traffic to protect, and state whether the ESP packets are in transport or tunnel mode, justifying your answer from what you can see in the capture. **[1 mark]**

Ans: The sender protects traffic according to a configured security policy, rather than protecting all traffic. The ESP packets are in transport mode because the original ipv4 header remains visible and ESP is carried as the IP Payload. The entire original IP packet has not been encapsulated inside a new IP header